Local and Global Retargeting Scope for MABEL:
Is the Mobile Base in the Follow Loop?

(`mabel.teleop_engine`, `mabel.whole_body_retargeting`)

MABEL Project — Control & Teleoperation

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Abstract

MABEL's Vision Pro teleoperation regains a two-mode retargeting split, on a different axis from the one retired in July. **LOCAL** (the default) puts the arms, hands, torso, lift and neck in the follow loop and leaves the mobile base to the operator's joystick; **GLOBAL** adds the base, which moves only once the operator leaves a comfort box in translation *and* in orientation. The two differ only in the base, which we verify rather than assert: right-arm tracking is $|r| = 1.000$ in both scopes with a lag difference of 0.2 ms. LOCAL commands exactly $0.0000\,\mathrm{m\,s^{-1}}$ of base motion through a walk-and-turn that GLOBAL answers with $0.8\,\mathrm{m\,s^{-1}}$, and a mid-session switch between them steps the base command by $0.0000\,\mathrm{m\,s^{-1}}$ because the base re-anchors at the live pose instead of catching up on displacement it deliberately ignored. Beyond the geometric box, base engagement is additionally scaled by how hard the arms are actually working, adapting the base-weighting insight of Haviland et al. to a cascade: at an identical 18 cm lean the base stays parked when the arms are comfortable and commands $0.57\,\mathrm{m\,s^{-1}}$ when they are near full extension. We also introduce an analytic three-frame facing cue — the head and both wrists agreeing — for which we find no published counterpart, and we pin single-implementation sim/real parity with a structural test suite after two divergences in one month.

Keywords: teleoperation, whole-body retargeting, mobile manipulation, redundancy resolution, sparse tracking, sim-to-real parity.

1 Overview

MABEL is a mobile manipulator: two 7-DoF arms with dexterous hands, a pitching torso, a vertical lift, a 3-DoF neck, and a swerve base. A Vision Pro streams three frames — the head and two wrists. Mapping three frames onto that many redundant DoFs raises one question above all others: *what should the base do?*

Answering it always leaves the arms solving against a moving root. So the mode split adopted here is not about the estimator's inputs but about the optimisation's variables:

- **LOCAL** (default) — arms, hands, torso, lift, neck follow the operator. The base is *not* in the loop; the operator drives it on the joystick, and the lift too. Deliberately the easier problem: a root that cannot move under the arm IK.
- **GLOBAL** — the base joins, taking the overflow past a comfort box in translation and past a deadband in orientation.

This is *not* the `frame` axis retired on 2026-07-28, which concerned which measurements the body-frame estimator was allowed to believe. That distinction matters enough that the wire documentation states it explicitly, so the two are never conflated again.

1.1 Relation to the literature

Surveying headset-driven mobile manipulation, the base/arm split is resolved in four ways, and *none* of them is the reach-boundary threshold one might assume: explicit operator channels (foot pedals in Mobile-TeleVision and HOMIE, a double-pinch in X-OP), a learned base agent (MoMa-Teleop, T-RO 2025), a learned hierarchical tracker (HiWET), or a continuous whole-body QP (Haviland et al., RA-L 2022). We adopt the operator's requested boundary form and then recover much of the QP's behaviour on top of it (§4).

Two further findings shaped this work. First, no robotics paper we found evaluates root/torso estimation error as a metric, though head-yaw-as-facing is measured at 14.5° mean error ($\sigma = 18.9$) in the graphics literature. Second, no published method uses an analytic agreement rule over head and both wrists to determine facing — the gap §6 fills.

2 The scope contract, measured

Fig. 1 and Table 1 give the definition of the modes as a measurement. A synthetic ARKit operator walks $0.35\,\mathrm{m\,s^{-1}}$ forward while turning $\pm 0.5\,\mathrm{rad}$.

Table 1: S1/S2. The modes differ in the base and *only* in the base.

scope	peak base command	right-arm $\lvert r\rvert$	right-arm lag
LOCAL	$0.0000\,\mathrm{m\,s^{-1}}$	1.000	$-5.9\,\mathrm{ms}$
GLOBAL	$0.8000\,\mathrm{m\,s^{-1}}$	1.000	$-6.1\,\mathrm{ms}$

The second and third columns are the load-bearing ones. If the scopes differed in how the *arms* track, the modes would not be what the documentation claims and an operator could not switch mid-task without relearning the feel. They do not: $\Delta|r| = 0.000$ and $\Delta\mathrm{lag} = 0.2\,\mathrm{ms}$.

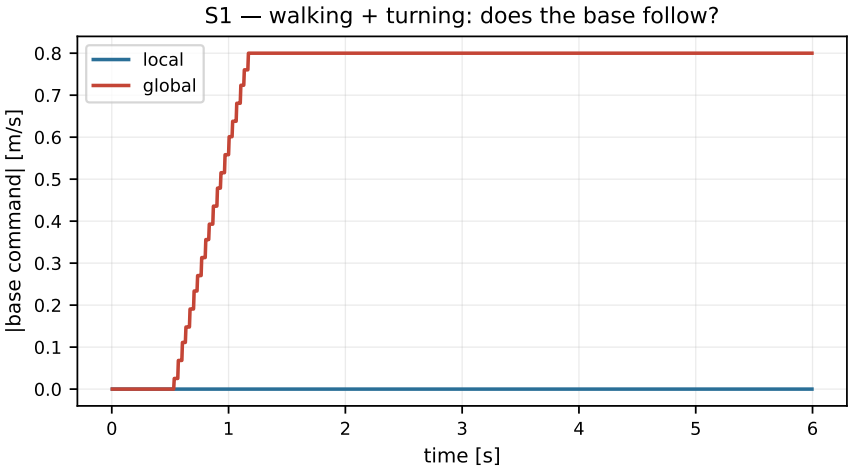


Figure 1: S1 — base command through a walk-and-turn, by scope.

3 Boundaries: declared versus effective

The base engages on overflow past a comfort box. Sweeping the operator's lean and body heading (Fig. 2) shows where it *actually* engages:

Table 2: S3. Effective boundaries exceed the declared constants.

axis	declared	measured engagement
translation	0.18 m	between 0.20 and 0.30 m
orientation	15°	between 20 and 34°

Both engage *later* than the constant names suggest, because the deadband acts on an error that then passes through a proportional controller and the Layer-1 motion-model ramp before any velocity is realised. The constant is a threshold on the error, not on the operator's displacement. This is the sort of gap that stays invisible until measured, and the numbers above are what should be retuned against if engagement is wanted at the nominal figure.

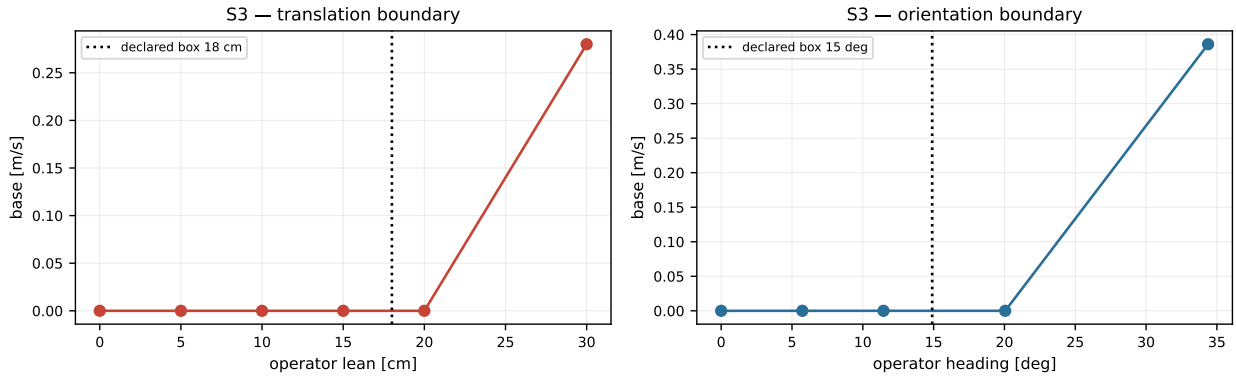


Figure 2: S3 — base engagement versus operator lean (left) and body heading (right), against the declared boxes.

The orientation sweep required care: turning only the *head* is a look-around, which the body-frame estimator is built to reject, and the first version of this experiment measured 0.000 at every angle — testing the estimator's rejection rather than the base's boundary. It now turns the head and the hands together.

4 Reach-aware engagement

A geometric box knows where the *head* is. It cannot express "the arms have this covered, stay put", because it never sees the arms. Haviland et al. (RA-L 2022) solve the equivalent problem inside a single whole-body QP by weighting the base DoFs as $\lambda_{\text{base}} = 1/\|e\|$: cheap when the end-effector error is large, expensive when it is small, so the base takes gross motion and the arm fine motion with no threshold at all (2–3× faster than plan-base-then-move-arm; 35 versus 71 failures in 1000 reaching tasks).

MABEL's base is a cascade rather than one QP, so we take the insight rather than the formulation. The box overflow is scaled by a *reach demand* $\rho \in [0, 1]$, the larger of the two sides' extension

fractions

$$\rho_{\text{side}} = \frac{\|w_{\text{side}} - s_{\text{side}}\|}{L_{\text{side}}}, \qquad \text{assist} = 1 + k \ \text{ramp}(\rho;\ \rho_{\text{free}}, \rho_{\text{full}}), \tag{1}$$

with w the operator's wrist, s their estimated shoulder and L their own arm length. Using the operator's *own* anthropometry matters: a fixed metric threshold reads a short-armed operator as permanently stretched, which is exactly the failure mode reported for MINK-IK in the SEW-Mimic user study. ρ is zero until the shoulders are estimated — no evidence, no assist.

Table 3: S6. Identical 18 cm lean and identical head displacement; only the arms' extension differs.

operator's arms	assist off	assist on
comfortable ($\rho \approx 0.5$)	$0.0000\,\text{m}\,\text{s}^{-1}$	$0.0000\,\text{m}\,\text{s}^{-1}$
stretched ($\rho \approx 1.0$)	$0.0000\,\text{m}\,\text{s}^{-1}$	$0.5733\,\text{m}\,\text{s}^{-1}$

`MABEL_BASE_REACH_ASSIST=0` restores the pure geometric box, so the two are A/B-able on the robot.

5 Laziness of the slow DoFs

The torso and lift extend reach; they should not mirror sway. Soft deadbands (6 cm of lean, 5 cm of head rise) shift the response toward zero rather than clipping it, so crossing the threshold never steps. Measured, the torso is silent at 2, 5 and 10 cm of sway and moves 3.14° at 20 cm.

No published work gives numeric deadband or hysteresis values for slow-DoF recruitment; the evidenced alternative is asymmetric posture regularisation (AMO reports 8× better height tracking, $0.117 \rightarrow 0.015$). The deadband form here is the operator's requested behaviour, and a sweep of the two constants against that alternative is open work.

6 Three-frame forward consensus

The operator's definition: *both hands facing forward and the head facing forward* is what a person naturally does when squared up. Previously the facing datum was head yaw alone — the estimator measured at 14.5° mean error, $\sigma = 18.9$ (Ponton et al., CGF 2022); a learned net over all three devices reaches 5.4°, $\sigma = 7.7$. We find no published *analytic* agreement rule over head and both wrists.

The obstacle is convention: a wrist frame's yaw is not "the direction the palm faces" in the head's convention, and guessing ARKit's hand-anchor layout wrong silently inverts the cue. Nothing is assumed. At Set Forward — where the operator is squared up *by declaration* — each wrist's yaw offset from the head is recorded; afterwards facing is simply all three de-offset yaws agreeing, with confidence falling off with their spread.

The last row is the exact false positive that sinks a head-only datum.

Closing a drift-correction loop on this cue is implemented but ships **disabled** (`FACING_CORRECT_ENABLED = False`): enabling it measurably regressed the estimator's canonical turn cases, which are pinned to fractions of a degree. It should be enabled only once its rate has been tuned against those same cases.

Table 4: Consensus behaviour, with arbitrary $+0.9/-0.6$ rad wrist offsets in play throughout.

operator state	confidence	heading
squared up	1.00	0.0°
30° whole-body turn	1.00	29.8°
one hand across the body	0.00	—
look-around (head turns, hands do not)	0.00	—

7 Bumpless mode transition

The base tracker integrates head displacement from an anchor, so a base that sat out of the loop while the operator walked has accumulated an arbitrarily large offset. Switching it in without re-anchoring commands a sprint to catch up on motion it deliberately ignored. `set_base_in_loop` re-anchors at the live pose instead: 1.2 m of displacement accrued in LOCAL commands $0.000\,\mathrm{m\,s^{-1}}$ on the switch, and a mid-session switch steps the base command by $0.0000\,\mathrm{m\,s^{-1}}$ (Fig. 3).



Figure 3: S5 — base command across a mid-session LOCAL→GLOBAL switch.

8 One controller for sim and robot

The same `TeleopEngine` inside the same `MabelTeleop` facade serves both, with `make_plant` as the sole fork: `MuJoCoPlant` owns physics in simulation, `ROSPlant` is pure transport on the robot and the engine's MuJoCo instance becomes a kinematic shadow. This had drifted twice in one month, each time producing the same confusing symptom — a fix that "does nothing" because the robot never had it — so it is now a test suite rather than a convention: the facade takes no sim/real parameter, neither engine nor facade branches on the dispatch target, the fork lives in `server/` and never in `controller/`, scope behaviour is plant-independent, the legacy scripts path is still a re-export shim rather than a second engine, and hyperparameters carry no built-in target branch.

The branch check tokenises rather than greps: these modules discuss the real robot at length in their docstrings, correctly, and a naive text search flags all of that prose — as the first version did, on a facade docstring.

What no test can enforce, and is therefore stated rather than implied: that a given *host* has pulled a given commit. That is a deployment fact, checked with `git log` on the robot.

9 Implementation and usage

- `controller/experiments/exp_retarget_scope.py` — S1–S5; regenerates every figure here.
- `controller/tests/test_sim_real_parity.py` — the 7 structural parity tests.
- `controller/tests/test_retarget_modes.py` — the scope contract at facade level, including that LOCAL is the default.
- Wire: `scope` on `control_mode`, additive and orthogonal to `region`; absent or unrecognised stays LOCAL.
- App: the LOCAL | GLOBAL segment sits beside Set Forward in the top bar, `MabelTeleopTests/RetargetScopeTests`.

10 Limitations

- **No hardware verification.** Every number is synthetic-operator and ideal-plant. Nothing here has run on the robot.
- **The declared constants are not the effective boundaries** (Table 2); they have not been retuned to close that gap.
- **Reach assist is unvalidated against the QP alternative.** It reproduces the qualitative behaviour of Haviland's weighting in a cascade, but the comparison against an actual whole-body QP has not been run.
- **Drift correction ships disabled** (§6).
- **The laziness deadbands are untuned** — one operating point, no sweep, and no comparison against posture regularisation.
- **Latency in the two scopes is not separately measured.** The engine's phase lag was measured at ≤ 12 ms overall (`vp_bimanual_latency.tex`) and S2 shows the scopes agree to 0.2 ms, but no end-to-end wire measurement exists per scope.

11 References

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